OPEN, CLOSED, LOCALLY CLOSED, CONSTRUCTIBLE AND LOCALLY UNION-OF-LOCALLY-CLOSED SUBSETS OF A JACOBSON SPACE ARE JACOBSON

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Allowed standard prerequisites (editorial).

- The definition of a Jacobson space through the property that every locally closed subset has a closed point, and the notion of a constructible set. Set up in the author's sections on Jacobson spaces and on constructible sets, before the counted statement. Fixes the property to be inherited and the list of subset types for which it is claimed.

Lemma 0.1. *Let X be Jacobson. The following types of subsets $T \subset X$ are Jacobson:*

(1) *Open subspaces.*
(2) *Closed subspaces.*
(3) *Locally closed subspaces.*
(4) *Unions of locally closed subspaces.*
(5) *Constructible sets.*
(6) *Any subset $T \subset X$ which locally on X is a union of locally closed subsets.*

In each of these cases closed points of T are closed in X.

Proof. Let X_0 be the set of closed points of X. For any subset $T \subset X$ we let $(*)$ denote the property:

(*) Every nonempty locally closed subset of T has a point closed in X.

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Note that always $X_0 \cap T \subset T_0$. Hence property $(*)$ implies that T is Jacobson. In addition it clearly implies that every closed point of T is closed in X.

Suppose that $T = \bigcup_i T_i$ with T_i locally closed in X. Take $A \subset T$ a locally closed nonempty subset in T, then there exists a T_i such that $A \cap T_i$ is nonempty, it is locally closed in T_i and so in X. As X is Jacobson A has a point closed in X. $\square$